Lab Four

Jackson Delk

1. Analysis

The result depends on the arguments and on what the two dependencies return, so the dependency results are treated as inputs too. Both `updateStatus` and `sendEmail` can fail two different ways: they can return false (then `checkoutResource` throws its own exception), or they can throw the exception themselves (it just propagates up). Those are different code paths, so each gets its own partition.

Input	Partitions
resourceId	null (invalid), any non-null UUID (valid)
isResourceAvailable(id)	returns true, returns false
updateStatus(id, false)	returns true, returns false, throws DatabaseFailureException
sendEmail(email, msg)	returns true, returns false, throws EmailFailureException
memberEmail	any String (the spec gives no rules for it, so one partition)

Output	Partitions
Return value	true, false
Exception	DatabaseFailureException, EmailFailureException, none
Side effects	updateStatus called / not called, sendEmail called / not called

Boundary value analysis doesn't apply here since none of the inputs are ordered ranges. A UUID is either null or it isn't, and availability is just true or false. Since the result depends on combinations of conditions, a decision table fits better than plain EP.

2. Test Coverage Items (Equivalence Partitions)

TCI	Input / Output	Partition
EP1	resourceId	null
EP2	resourceId	non-null UUID
EP3	isResourceAvailable	returns true
EP4	isResourceAvailable	returns false
EP5	updateStatus	returns true
EP6	updateStatus	returns false
EP7	updateStatus	throws DatabaseFailureException
EP8	sendEmail	returns true
EP9	sendEmail	returns false
EP10	sendEmail	throws EmailFailureException
EP11	return value	true
EP12	return value	false
EP13	exception	DatabaseFailureException thrown
EP14	exception	EmailFailureException thrown

3. Decision Table

	R1	R2	R3	R4	R5	R6	R7
Conditions							

	R1	R2	R3	R4	R5	R6	R7
resourceId not null	F	T	T	T	T	T	T
isResourceAvailable	–	F	T	T	T	T	T
updateStatus	–	–	false	throws	true	true	true
sendEmail	–	–	–	–	false	throws	true
Actions							
Returns	false	false	–	–	–	–	true
Exception thrown	none	none	DB	DB	Email	Email	none
updateStatus called	N	N	Y	Y	Y	Y	Y
sendEmail called	N	N	N	N	Y	Y	Y

“–” means don’t care (if the ID is null, nothing after it matters). DB = DatabaseFailureException, Email = EmailFailureException. The order matches the code: availability is checked first, then the status is updated, then the email is sent.

4. Test Cases

Each rule gets its own test, which also keeps every invalid partition in its own test so one error can’t hide another. Test data used throughout:

```
id = UUID.fromString("123e4567-e89b-12d3-a456-426614174000")
email = "test@email.com"
```

Test	Rule	TCIs	Inputs / Mock Setup	Expected Result
T1	R1	EP1, EP12	resourceId = null, email	Returns false. No calls to the repository or the email provider.
T2	R2	EP2, EP4, EP12	id, email; isResourceAvailable(id) returns false	Returns false. updateStatus and sendEmail never called.
T3	R3	EP2, EP3, EP6, EP13	id, email; available; updateStatus(id, false) returns false	Throws DatabaseFailureException. sendEmail never called.
T4	R4	EP2, EP3, EP7, EP13	id, email; available; updateStatus(id, false) throws DatabaseFailureException	Throws DatabaseFailureException (the mock’s exception propagates). sendEmail never called.
T5	R5	EP2, EP3, EP5, EP9, EP14	id, email; available; updateStatus returns true; sendEmail returns false	Throws EmailFailureException. updateStatus(id, false) was called once.
T6	R6	EP2, EP3, EP5, EP10, EP14	id, email; available; updateStatus returns true; sendEmail throws EmailFailureException	Throws EmailFailureException (the mock’s exception propagates). updateStatus(id, false) was called once.
T7	R7	EP2, EP3, EP5, EP8, EP11	id, email; available; updateStatus returns true; sendEmail returns true	Returns true. updateStatus(id, false) called once. sendEmail("jdoe@example.com", "Resource ID: 123e4567-e89b-12d3-a456-426614174000 checked out.") called once.

All 14 TCIs are covered by at least one test. The “never called” and “called once with” checks use Mockito’s `verify(...)`, `verify(..., never())` and `verifyNoInteractions(...)`. Only checking the return value isn’t enough, since a version that returned true without actually sending the email would still pass.